CODECHELLA MADRID 2026

Continuous Difference-in-Differences

From TWFE with a continuous dose to $ATT(d|d)$ and the ACRT

Scott Cunningham · Baylor University

CUNEF Auditorium · May 25–28, 2026

Continuous treatments are everywhere

When we say "DiD," we usually mean a *binary* treatment: $D_i \in \{0, 1\}$. But empirical work routinely uses DiD with a *continuous* dose:

- Minimum wages (*cents* of bite, not just "adopted")
- Abortion clinic closures (*kilometers* to the nearest clinic)
- Vaccination campaigns (*doses per 1,000*)
- Tariff cuts (*percentage-point reductions*)
- Pollution (μ*g/m*^{3})

The treatment isn't "on/off"; it has a magnitude. The question becomes "how much effect per unit of dose?" — not just "effect of treatment."

The classic praise for OLS with a continuous dose

> *The two-period regression estimator can be easily modified to allow for continuous, or at least non-binary, treatments.*

— Wooldridge (2005)

> *A second advantage of regression DiD is that it facilitates the study of policies other than those that can be described by a dummy.*

— Angrist & Pischke (2008)

The encouragement was: run the same TWFE regression, but make D a continuous variable. Easy.

The TWFE specification, with a continuous dose

The standard move is to write the same TWFE model, but let the treatment intensity vary:

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \beta\, D_i \cdot \text{Post}_t \;+\; \varepsilon_{it}$$

Where now:

- $D_i \in [0, \bar{d}]$ is a *dose* (kilometers, dollars, percentage points)
- Post_t is the usual post-period indicator
- β is read as "effect of one more unit of dose"

This is what almost every applied paper in the literature has used. What does β actually identify?

New work says: $\beta^{\textbf{twfe}}$ does not have a clean causal reading

Callaway, Goodman-Bacon, and Sant'Anna (2025), *"Difference-in-Differences with a Continuous Treatment"* (**CBS**).

The core claim. Under unrestricted heterogeneity in treatment effects across doses, the TWFE coefficient $\widehat{\beta}^{\text{twfe}}$ is a *weighted average* of dose-specific effects — with weights that:

- Are *not* non-negative.
- Are *not* the dose density.
- Can place mass on *differences* between doses that the researcher never intended to compare.

We unpack these weights on Lu & Yu (2015) next.

Plan for the lecture

Part 1. The TWFE continuous-dose specification, and its decomposition. (Lu & Yu 2015 / China WTO accession as our running example.)

Part 2. Potential outcomes and building blocks. ATT($d \mid d$), the ACRT, standard vs strong parallel trends.

Part 3. Estimation: dose-specific 2$\times$2's, overall ATT, bins, non-parametric smoothing. The `contdid` package.

We will spend more time on the decomposition than on estimation. Understanding why β^{twfe} has the structure it has is the foundation.

Lesson

Part 1:
The TWFE decomposition

Lu & Yu (2015) reported $\widehat{\beta}^{\text{twfe}} = -0.322$

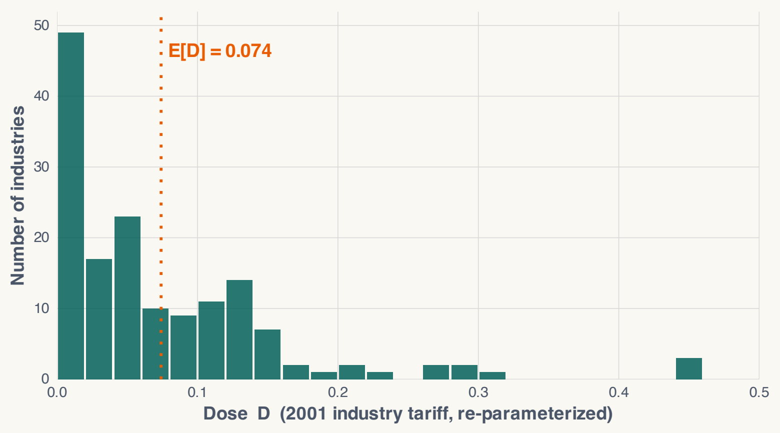
Lu & Yu (2015), *"Trade liberalization and markup dispersion: Evidence from China's WTO accession," AEJ: Applied Economics* 7(4): 221–253.

The result. A one-percentage-point cut in the import tariff facing industry i *decreased* markup dispersion by 0.322 log points. Single coefficient $\widehat{\beta}^{\text{twfe}} = -0.322$.

But $\widehat{\beta}^{\text{twfe}}$ is not the "effect of a tariff cut." It is a weighted average of dose-specific effects, with weights set by the dose distribution.

We will compute one of those weights by hand by the end of this section.

China's WTO accession: 155 industries, different tariff cuts



Tariff cut across 155 Chinese industries. ***Mean*** $\approx 7\%$; *range* 0 *to* $\sim 50\%$.

One coefficient $\widehat{\beta}^{\text{twfe}}$ carries the entire dose response

The Lu & Yu specification is the canonical continuous DiD:

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \beta^{\text{twfe}}\,(D_i \cdot \text{Post}_t) \;+\; \varepsilon_{it}$$

One coefficient. No dose-by-dose breakdown. No interaction with D_i^2 or with bins. Just β^{twfe}, the slope of Y on D in differenced units.

One coefficient, three questions:

1. Effect of any tariff cut at all?
2. Effect per percentage point of cut?
3. Effect at a particular dose, say 10%?

$\widehat{\beta}^{\text{twfe}}$ is supposed to answer all three at once. It cannot.

$\widehat{\beta}^{\text{twfe}}$ is a weighted average of dose-level effects

CBS show that the OLS coefficient can be written as:

$$\widehat{\beta}^{\text{twfe}} = \int_0^{\bar{d}} w(l) \cdot \Big(\text{effect at dose } l\Big)\, dl$$

This is pure algebra. No potential outcomes yet, no assumptions about identification — just what the regression *is*, mechanically, as a weighted sum.

Four different weights $w(l)$ can be motivated. Each is a real, derivable answer to "what is $\widehat{\beta}^{\text{twfe}}$?"

Next: write down the four weight formulas. Potential outcomes show up in Part 2.

The four weight formulas, all at once

$$\textbf{Level weight:} \quad w^{\text{lev}}(l) = \frac{(l - E[D]) \cdot f_D(l)}{\text{Var}(D)}$$

$$\textbf{Scaled level:} \quad w^{s}(l) = l \cdot \frac{(l - E[D]) \cdot f_D(l)}{\text{Var}(D)}$$

$$\textbf{ACRT weight:} \quad w^{\text{acrt}}(l) = \frac{(E[D \mid D \geq l] - E[D]) \cdot P(D \geq l)}{\text{Var}(D)}$$

$$\textbf{2}\times\textbf{2 weight:} \quad w^{2\times 2}(l, h) = \frac{(h - l)^2 \cdot f_D(l) \cdot f_D(h)}{\text{Var}(D)}$$

Same six features of the dose distribution; four ways to recombine them.

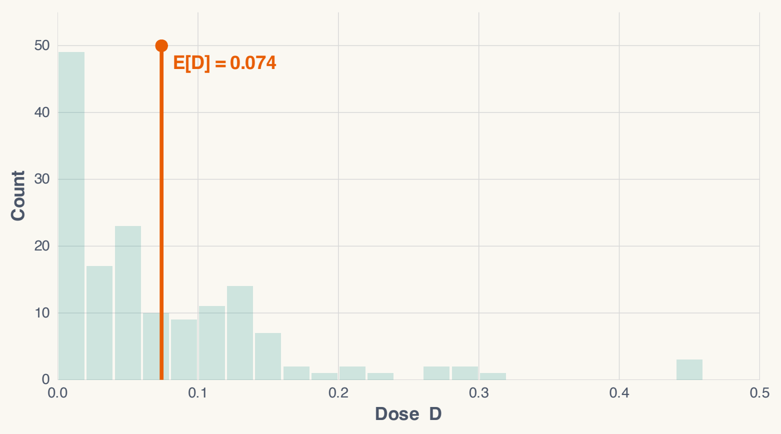
Six ingredients combine into the four weight formulas

Every weight formula is built from the same six features of the dose distribution:

1. $E[D]$ — the mean dose
2. $\text{Var}(D)$ — the variance
3. $f_D(l)$ — the density at dose l
4. $P(D \geq l)$ — the survival probability
5. $E[D \mid D \geq l]$ — the conditional mean above l
6. d_L — the minimum positive dose

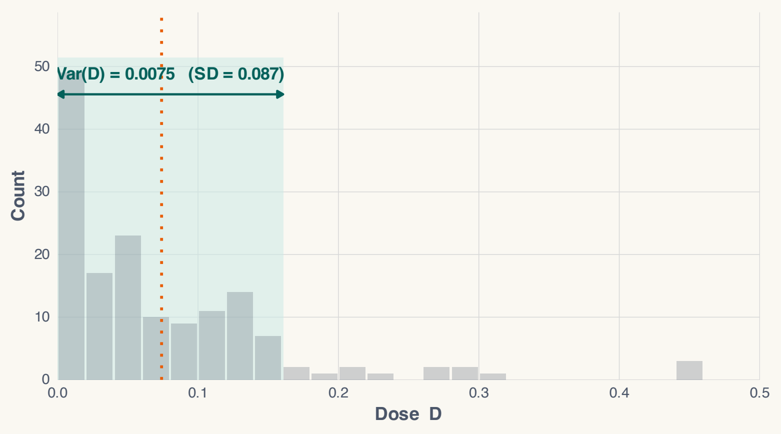
Same six numbers; four different recombinations.

Ingredient 1: $E[D]$ anchors every formula



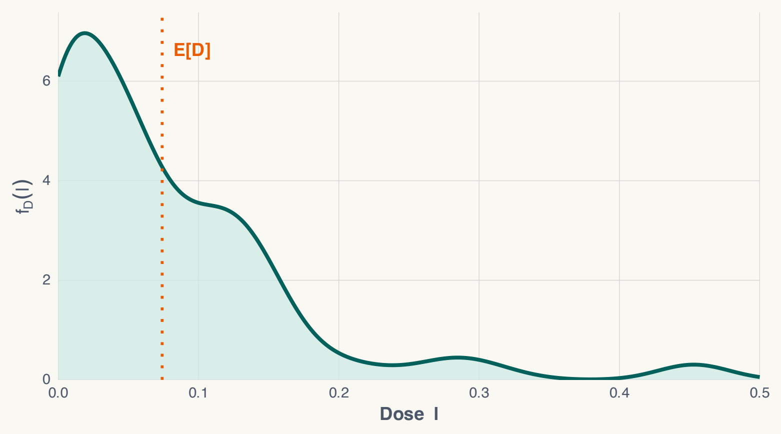
WTO mean dose $\approx 7.1\%$. The weights flip sign around this value.

Ingredient 2: Var(D) sets the scale



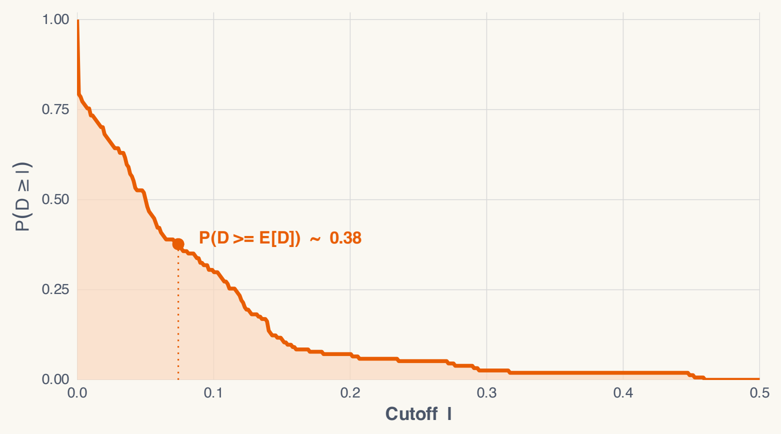
The variance shows up in the denominator of the level weight: a higher Var(D) makes the formula scale things down.

Ingredient 3: $f_D(l)$ — “where are the industries?”



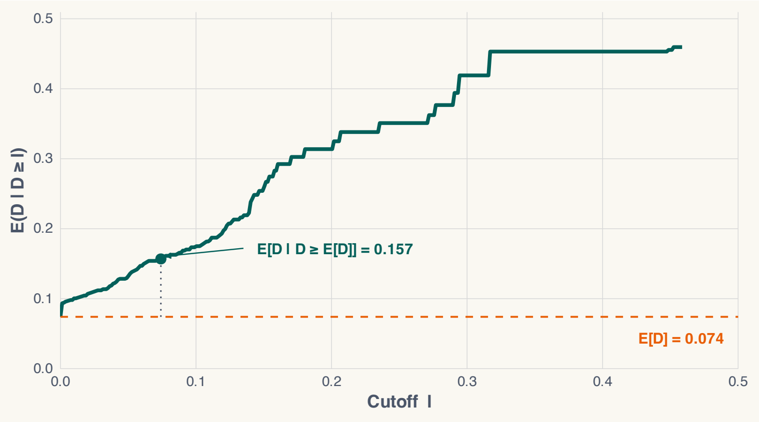
The density says where doses cluster. The ACRT weight uses f_D directly; the level weight does not.

Ingredient 4: $P(D \geq l)$ — “how many above the cutoff?”



The survival probability counts industries with dose $\geq l$. It enters the ACRT formula.

Ingredient 5: $E[D \mid D \geq l]$ tracks the upper tail



The conditional mean of dose above cutoff l. Shows up in the ACRT weight.

Ingredient 6: d_L, the minimum positive dose

The smallest non-zero dose in the data. In WTO it is about 0.05% (one industry whose 2001 tariff sat just above the 10% bound). Doses $l < d_L$ have weight zero — no industry experiences that small a cut.

Why we name it. Several formulas integrate "from d_L to $\bar{d}$." If you fail to use d_L you can put mass on doses no one ever received — a textbook source of misinterpretation.

All six ingredients are computed from the cross-section of doses alone — before the outcome is even loaded.

The regression never changes. The question does.

One TWFE regression. One $\widehat{\beta}^{\text{twfe}}$.

FWL lets you rewrite that single number as a weighted average — *four different ways*.

- Same $\widehat{\beta}$ each time.
- Different *weights* doing the averaging.
- Different things being *averaged over*.

Four rewrites. Four implicit questions. OLS doesn't tell you which one you asked.

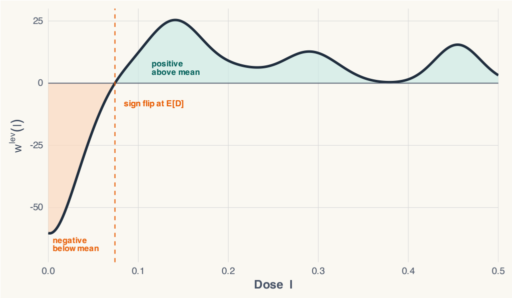
Four questions, four decompositions

The question	Target parameter	Decomposition
Q1. *Effect of the program.* Industry at dose d vs. untreated.	$\text{ATT}(d \mid d)$	level weight w^{lev}
Q2. *Effect per unit of dose.*	$\dfrac{\text{ATT}(d \mid d)}{d}$	scaled-level w^{s}
Q3. *Marginal effect.* Tiny nudge at dose d.	$\text{ACRT}(d)$	ACRT weight w^{acrt}
Q4. *Two treated industries.* Dose l vs. dose h.	$\text{ATT}(h \mid h, l)$	2×2 weight $w^{2\times 2}$

All four rewrites equal the same $\widehat{\beta}$. The number doesn't tell you which one you wanted.

Weight #1: the level weight $w^{\text{lev}}(l)$

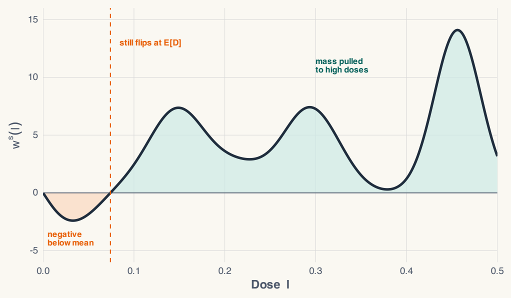
$$w^{\text{lev}}(l) = \frac{(l - E[D]) \cdot f_D(l)}{\text{Var}(D)}$$



Below-mean doses get negative weight; above-mean doses get positive.

Weight #2: the scaled-level weight $w^s(l)$

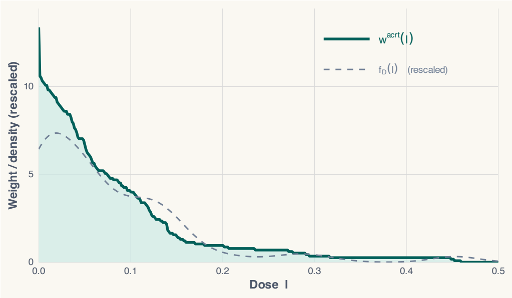
$$w^s(l) \;=\; l \cdot \frac{(l - E[D]) \cdot f_D(l)}{\text{Var}(D)}$$



Mass pulled to high doses; sign flip remains. Integrates to 1.

Weight #3: the ACRT weight $w^{\text{acrt}}(l)$

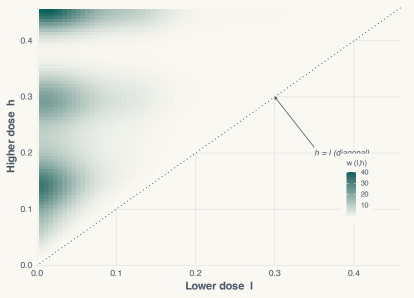
$$w^{\text{acrt}}(l) = \frac{(E[D \mid D \geq l] - E[D]) \cdot P(D \geq l)}{\text{Var}(D)}$$



Non-negative, integrates to 1. Shape differs from the density f_D.

Weight #4: the 2×2 weight $w^{2\times 2}(l, h)$

$$w^{2\times 2}(l,h) = \frac{(h-l)^2 \cdot f_D(l) \cdot f_D(h)}{\mathrm{Var}(D)}$$



Far-apart pairs get the most weight; $(h-l)^2$ shows up visually.

Same dose distribution, four different stories

Four overlaid weights, same distribution. $\widehat{\beta}^{twfe}$ *is all of them at once.*

Let's verify it with data

We've described the weights. Now let's actually compute them and rebuild $\widehat{\beta}^{\text{twfe}}$ by hand.

Three steps:

1. Run OLS of $\Delta \ln$ Theil on dose. Save $\widehat{\beta}_{\text{OLS}}$.
2. Compute per-industry level weights $W_i = \dfrac{D_i - \bar{D}}{\sum_j (D_j - \bar{D})^2}$.
3. Form $\sum_i W_i \cdot \Delta \ln \text{Theil}_i$ and compare.

If FWL is right, the two numbers are identical.

```
Labs/China-WTO/twfe_reconstruction.R # R version
Labs/China-WTO/twfe_reconstruction.do # Stata version
```

The reconstruction in Stata

```
* (1) beta_hat from OLS
reg Delta_ln_theil dose
local beta_ols = _b[dose]

* (2) Per-industry level weights W_i = (D_i - Dbar) / SS_D
egen D_bar = mean(dose)
gen W_num = dose - D_bar
egen SS_D = total(W_num^2)
gen W = W_num / SS_D

* (3) Reconstruct beta_hat by hand
gen contrib = W * Delta_ln_theil
egen beta_manual = total(contrib)
```

R companion at `Labs/China-WTO/twfe_reconstruction.{R,do}`

$\widehat{\beta}^{\text{twfe}}$ is the weighted average. Exactly.

Quantity	Value
$\widehat{\beta}_{\text{OLS}}$ from `reg Delta_ln_theil dose`	-0.493524
$\sum_i W_i \cdot \Delta \ln \text{Theil}_i$ (weighted sum)	-0.493524
$\lvert\widehat{\beta}_{\text{OLS}} - \sum_i W_i \cdot \Delta \ln \text{Theil}_i\rvert$	5.55×10^{-17}

Numerically identical to machine precision.

The decomposition *is* the regression.

Negative weights aren't a bug — they're the price

- **Q1, Q2 (level, scaled-level).** Clean comparison: treated dose d vs. untreated. Weights forced to sum to zero $\Rightarrow$ below-mean doses get *negative* weights.
- **Q4 (2$\times$2).** Weights non-negative, integrate to 1 — but compare two *treated* industries to each other. A forbidden contrast.
- **Q3 (ACRT).** In between: non-negative weights, but the answer is a derivative, not a level.

Clean comparisons cost you negative weights. Tidy weights cost you forbidden comparisons. Pick your poison.

The takeaway from the decomposition

$\widehat{\beta}^{\textbf{twfe}}$ **is well-defined.** It is a sample statistic. But:

- It is *not* the average dose effect (that would be $\int f_D(l)\,\mathrm{ATT}(l \mid l)\,dl$).
- It is *not* the ACRT (that uses w^{acrt}, not w^{lev}).
- It *is* a weighted combination, with sign-flipping weights, of dose-specific effects.

The path forward. Pick the target parameter first. Estimate that parameter. Don't read $\widehat{\beta}^{\text{twfe}}$ as "the" dose effect.

Part 2 defines those target parameters. Part 3 estimates them.

Play with the decomposition: the `baconplus` Shiny app

Interactive companion to everything we just did:

https://scunning1975.github.io/baconplus/

What it does. Slider-based exploration of the TWFE decomposition on 155 simulated industries with a tariff-cut dose mimicking Lu & Yu (2015):

- Move the kernel bandwidth and watch $f_D(l)$ re-shape.
- See the level weight $w^{\text{lev}}(l)$, the scaled-level $w^s(l)$, and the ACRT weight $w^{\text{acrt}}(l)$ rebuild in real time.
- Reassemble $\widehat{\beta}^{\text{twfe}}$ from any of the four weight formulas.

Source: github.com/scunning1975/baconplus — CBS continuous-DiD TWFE decomposition implementation.

LESSON

Part 2: Potential outcomes and building blocks

ATT, but for a continuous dose

Recall the binary ATT:

$$\text{ATT} \;=\; \mathbb{E}\Big[\,Y_{it}(1) - Y_{it}(0) \mid D_i = 1,\ \text{Post}_t\,\Big]$$

With a continuous dose we need a richer object. Each unit has a potential outcome *at every possible dose*: $Y_{it}(d)$ for $d \in [0, \bar{d}]$.

Two natural parameters:

1. Effect of *being at* dose d vs being at zero. ("ATT at dose d")
2. Effect of *moving* from dose d to dose $d + \Delta$. ("causal response")

These are not the same thing.

Dosage parameters give us a multitude of ATTs

ATT for a given dose

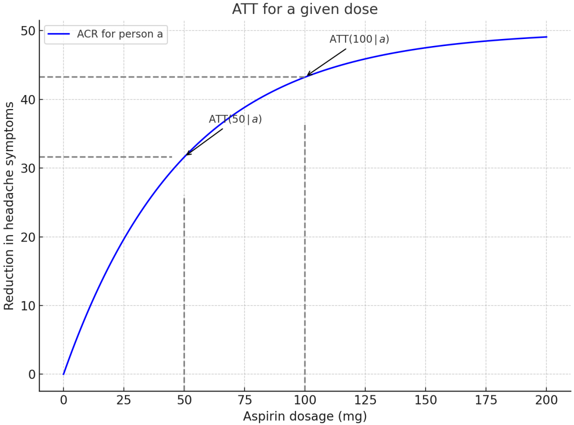
$$\text{ATT}(d \mid d) \;=\; \mathbb{E}\Big[\,Y_{it}(d) - Y_{it}(0) \mid D_i = d\,\Big]$$

Read it as: "The ATT of dose d for the group of units that actually *received* dose d, using zero dose as the counterfactual."

Example. What is the effect of being 100 km from a hospital, compared to being 0 km? Comparison is no-dose; population is the units actually 100 km away.

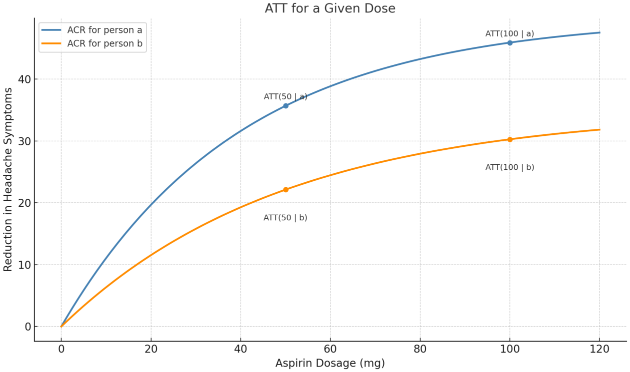
One ATT per dose; the full set is the dose-response curve.

ATT for a given dose: the aspirin example (1)



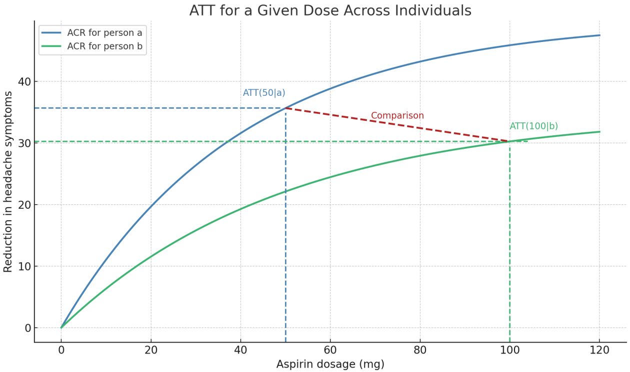
One person, one outcome (headache pain). At 50 mg, the effect is 31; at 100 mg, the effect is 44.

ATT for a given dose: the aspirin example (2)



ATT$(50 \mid 50)$ *vs ATT*$(100 \mid 100)$*: two separate ATT numbers, each comparing dose* d *to zero dose.*

Moving along the dosage is *not* the ATT



The difference between ATT(100) *and ATT*(50) *is a slope, not an ATT. It's a marginal effect along the dose response curve.*

The Average Causal Response on the Treated — ACRT

ACRT

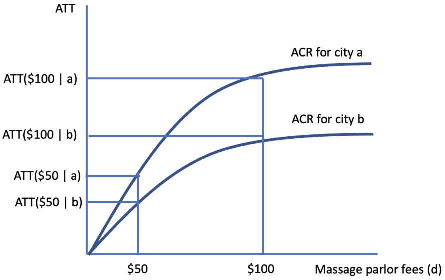
$$\text{ACRT}(d) \;=\; \frac{\partial}{\partial d}\,\mathbb{E}\Big[\,Y_{it}(d) \mid D_i = d\,\Big]$$

Read it as: "How much does the outcome change if I bump dose up by a tiny amount, evaluated at units actually receiving that dose?"

A slope, not a level. A marginal effect: mg per mg, km per km, point per point.

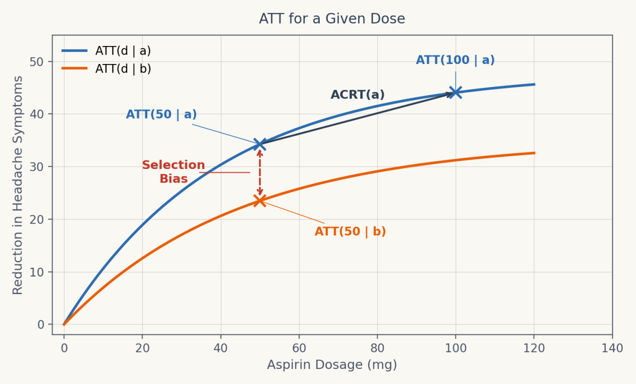
Angrist & Imbens (1995) defined the IV analog, ACR.

ACRT as a derivative



ACRT(d) is the slope of the ATT-dose curve at dose d — the marginal effect of one more unit of treatment.

Two different parameters, two different questions



ATT$(d \mid d)$ *is a level; ACRT*(d) *is a slope. Different objects, often confused.*

Identification: assumptions for ATT($d \mid d$)

A1. No Anticipation. Treatment doesn't move outcomes before it begins.

$$Y_{it}(d) \;=\; Y_{it}(0) \quad \text{for all } t < g_i$$

A2. Standard Parallel Trends (PT). The trend in $Y_{it}(0)$ is the same for treated units regardless of their realized dose:

$$\mathbb{E}\Big[\Delta Y_{it}(0) \mid D_i = d\Big] \;=\; \mathbb{E}\Big[\Delta Y_{it}(0) \mid D_i = 0\Big]$$

This gives identification of ATT$(d \mid d)$ for each $d > 0$. The level-effect dose-response curve, against the no-dose counterfactual.

Standard PT is the natural extension of binary-PT to a continuous setting.

No Anticipation $\Rightarrow$ baseline is $Y(0)$ for everyone

A1 says: pre-treatment potential outcomes do not depend on the future dose.

Combined with the data structure (no one is observationally treated yet in the pre-period), this gives:

$$Y_{i,\text{pre}} \;=\; Y_{i,\text{pre}}(0) \quad \text{for every unit } i, \text{ including those who get } D_i > 0 \text{ later.}$$

Why this matters. Every DiD comparison the model can express is mechanically a 2×2:

- *before* = the common untreated trajectory (both groups).
- *after* = treated units show $Y(d)$; control units stay on $Y(0)$.

The $(D \times$ Post$)$ interaction in TWFE is the mechanical consequence.

Identification: ACRT requires more — "Strong PT"

A2′. Strong Parallel Trends. The *counterfactual* trend at *every* dose is the same across treatment groups:

$$\mathbb{E}\Big[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d'\Big] \;=\; \mathbb{E}\Big[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d\Big]$$

In words. The treatment effect of dose d on a unit that actually received dose d equals the treatment effect of dose d on a unit that received dose d'.

This rules out selection on gains across doses. Strong PT is a substantial restriction; standard PT alone does not yield the ACRT.

Under standard PT we identify the level curve; under strong PT we identify the slope as the ACRT.

Why strong PT is stronger: a thought experiment

Suppose two industries: A chose a 10% tariff cut; B chose a 30% cut. Dose was not random.

Standard PT says: A and B would have had the same *baseline* trend, absent any cut.

Strong PT says, in addition: *if A and B both faced a 20% cut*, they'd respond identically.

Strong PT rules out selection on dose-response gains.

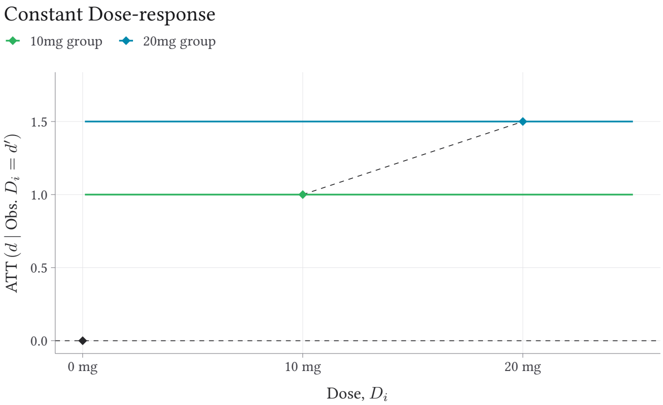
A discrete multi-valued example: pills, groups, DiD

To see how DiD operates on dose groups, drop continuous and consider a *multi-valued* discrete dose: $D \in \{0, 1, 2\}$.

Setting. Three groups of patients: untreated ($D = 0$), one pill ($D = 1$), two pills ($D = 2$). The discrete case bridges binary and continuous — same ideas, finite groups.

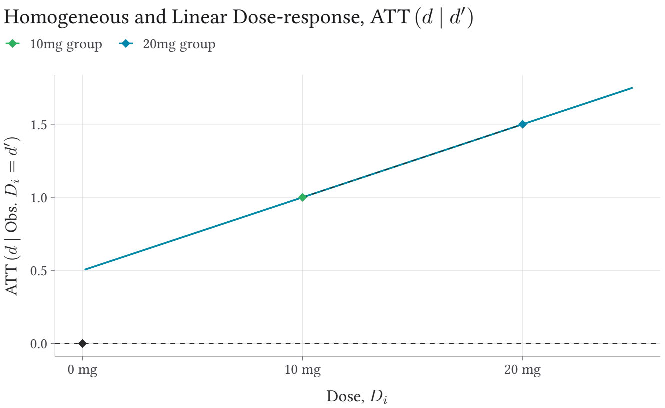
Next slides. Dose-by-dose ATTs, first-differences, the DiD, what's actually identified.

Discrete: raw outcomes by dose group



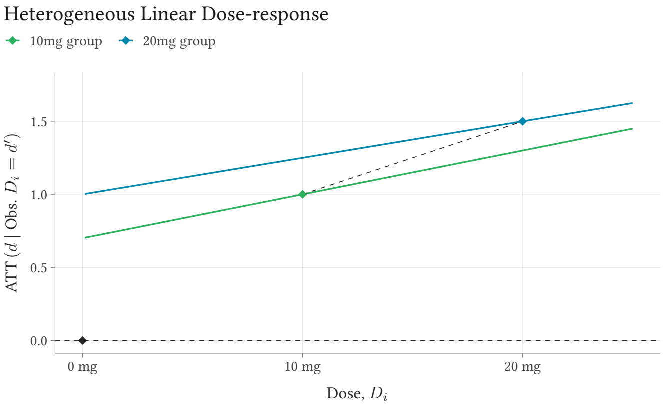
Three dose groups (0, 1, 2 pills). Their pre-period levels and trends are all parallel — this is the no-treatment-effect benchmark.

Discrete: with treatment effects (linear, homogeneous)



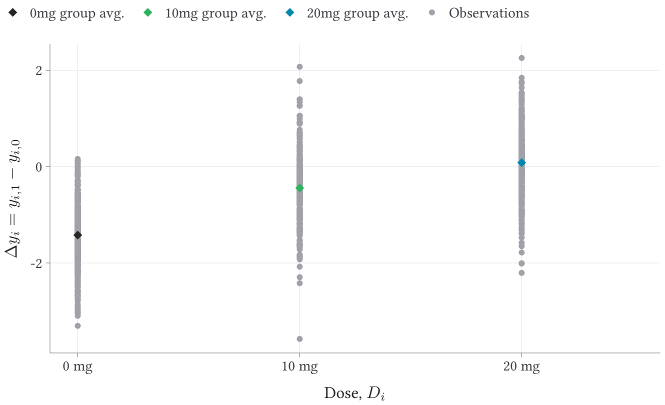
Dose-1 group lifts by δ; dose-2 group lifts by 2δ. Effects linear in dose; everyone responds the same way.

Discrete: with heterogeneity in the dose response



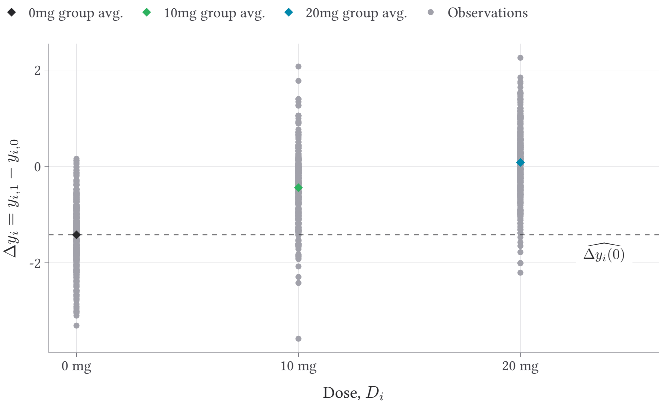
Same average response, but the dose-1 group reacts more than dose-2's first-pill response would have predicted. Heterogeneity in returns to dose.

Discrete: first differences make the comparison visible



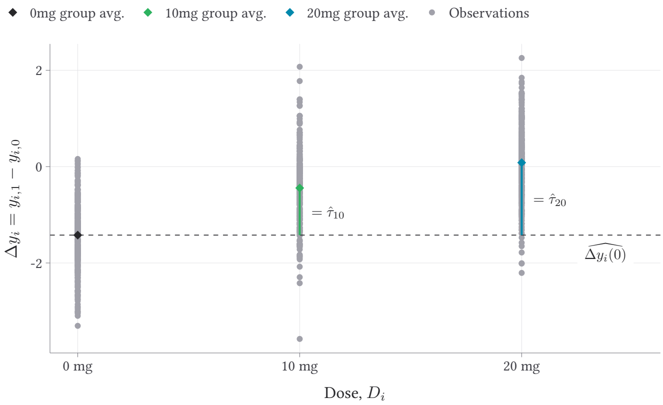
ΔY for each dose group. A clean way to read the pre-period (should be flat) vs the post-period (the treatment effect).

Discrete: the implied counterfactual



The dashed lines show what dose-1 and dose-2 would have looked like absent treatment — constructed under standard PT.

Discrete: the DiD estimates, dose by dose



Each treated dose-group's DiD versus the never-treated: $\widehat{ATT}(1 \mid 1)$ and $\widehat{ATT}(2 \mid 2)$. Two separate point estimates, one per dose.

Recap of Part 2

- Two target parameters: **ATT($d \mid d$)** (level) and **ACRT(d)** (slope).
- Standard PT identifies the level curve.
- Strong PT — ruling out selection on dose response — identifies the ACRT.
- Discrete multi-valued case gives finite-group intuition; continuous extends the same logic to a curve.

Part 3 estimates these objects, with three options: overall ATT, bins, and non-parametric smoothing.

LESSON

Part 3: Estimation

Backwards engineering vs. Forwards engineering

Backwards

- Run TWFE; let OLS hand you a number.
- FWL: that number is a weighted average of dose effects.
- Weights come from Var$(D \mid$ controls$)$.
- Some weights are **negative**; they shift across the dose distribution.
- No causal parameter in sight.

$\longrightarrow$

Forwards

- Name the target: ATT$(d \mid d)$ or ACRT(d).
- State assumptions you'll defend.
- Build an estimator that's right on average under those assumptions.
- Meaning of the estimate is known *before* you run it.

CBS is forwards. TWFE with continuous D is backwards.

A familiar precedent: Wald IV and the LATE

The ATE had long been the target parameter.

- Imbens & Angrist (1994) started with an estimator — Wald IV — and worked out what it actually identifies under heterogeneous effects: the **LATE**, the average effect among compliers.
- They met opposition. The LATE wasn't the ATE; critics argued you couldn't pick it ex ante.
- Eventually the field came around. Today: name the target (LATE, ATT, ...) *first*, then build the estimator.
- We're doing the same move here. TWFE on continuous D isn't the ATT. Pick ATT$(d \mid d)$ or ACRT(d); then estimate.

`youtube.com/watch?v=QnwTdd-7BvY`

Three estimation options for ATT($d \mid d$)

Given continuous D, three practical strategies:

1. **Overall ATT.** Average $\text{ATT}(d \mid d)$ across the treated. One number.
2. **Bins.** Partition the dose into intervals; estimate one ATT per bin. A coarse dose-response curve.
3. **Non-parametric smoothing.** A continuous dose-response function $\widehat{\text{ATT}}(\cdot)$ via kernels or splines.

Each one is a different point on the bias-variance-interpretability trade-off.

Option 1: Overall ATT — one number summary

Definition.

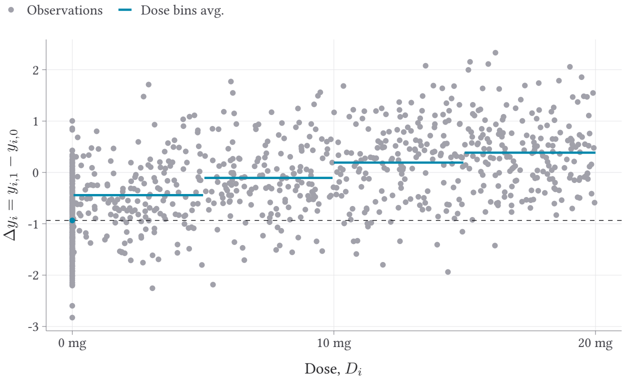
$$\overline{\text{ATT}} = \mathbb{E}\Big[\text{ATT}(D_i \mid D_i) \mid D_i > 0\Big]$$

The average of the dose-specific ATTs, taken over the treated dose distribution. The weight is the density f_D — not the level weight or any other formula.

When to use it. When the audience wants a single headline ATT. When the dose response is not the object of interest. When precision matters more than dose-by-dose detail.

This is what most empirical papers want, but it is not what TWFE delivers.

Option 2: Bins — a coarse dose-response curve



Partition dose into intervals. Estimate one ATT per bin. Step-function dose response.

Pros and cons of bins

Pros.

- Easy to estimate; nothing more than a series of 2×2's.
- Easy to communicate: "effect at 5–10% cut was X."
- Bins can be chosen with policy thresholds in mind.
- Identified under *standard* parallel trends — no need for Strong PT.

Cons.

- Inside a bin you average over a range of doses — losing within-bin variation.
- Choice of bin width is a discretion knob.
- No standard error on the dose-response curve as a continuous object.

Often the right first cut. Rarely the final analysis.

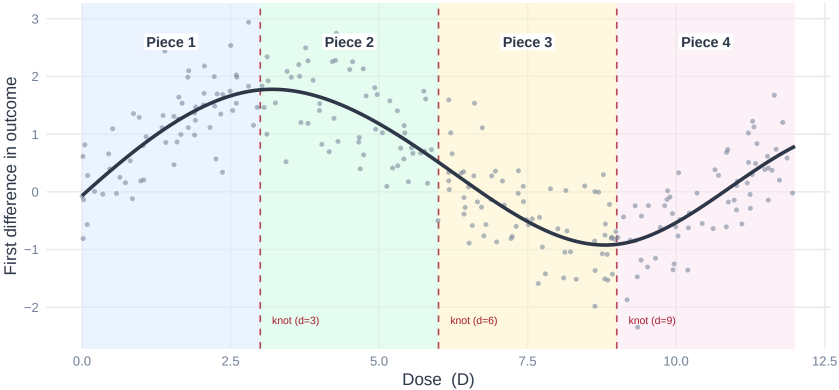
Why not just shrink the bins?

If bins of width h work, why not let $h \to 0$ and recover a curve?

- Each bin holds fewer observations as h shrinks. The bin average gets noisier.
- Variance grows faster than bias falls. The estimator stops converging.
- Inside a bin the slope is zero. You can't read ACRT(d) off a step function.
- Step functions don't have derivatives. You can't take limits.

Fine bins are not the answer. We need a smooth curve.

Splines: the curve is made of pieces



Simulated data. Three interior knots (dashed red) create 4 polynomial pieces. The curve is continuous, smooth, and differentiable at every knot.

3 knots divide the dose range into 4 regions. Inside each region: one cubic polynomial.
The curve is forced to match in value, slope, and curvature at every knot.

Fitting a spline in Stata: `mkspline`

`mkspline` generates the basis columns; `reg` does the rest.

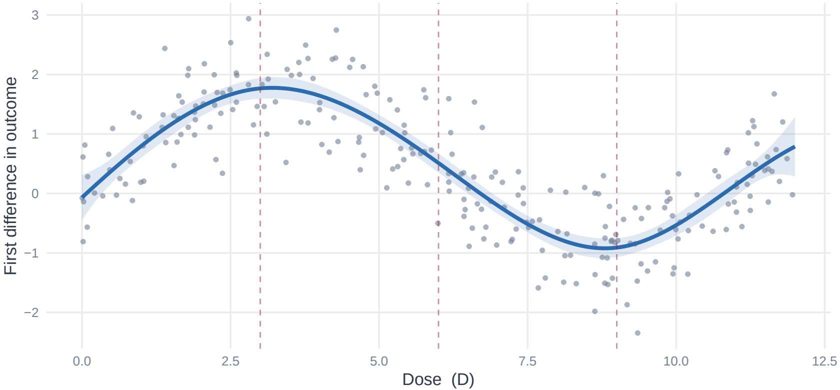
```
* 1. Make the basis (4 quantile knots = 3 interior knots = 4 pieces)
mkspline spl = dose, nknots(4) cubic displayknots

* 2. Regress the first-differenced outcome on the basis
reg delta_y spl*

* 3. Fitted values and plot
predict yhat, xb
twoway (scatter delta_y dose, mcolor(gs10) msize(small)) ///
       (line yhat dose, sort lcolor(navy) lwidth(medthick)), ///
       xtitle("Dose") ytitle("First difference") legend(off)
```

`nknots(4)` places knots at dose quantiles. `displayknots` prints their values so you can report them.

What the spline estimates



Fitted cubic spline (blue line) with 95% pointwise confidence band. Dashed lines mark the 3 interior knots.

The `predict yhat` line from the regression above. One smooth curve through the scatter. Pointwise confidence band from `predict se, stdp` and `twoway rarea`.

Kernel regression: another way to smooth

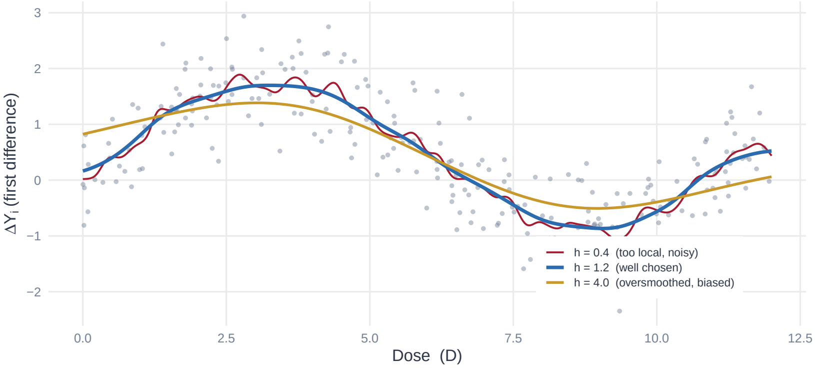
At each dose d, average outcomes from nearby observations, weighted by a kernel:

$$\widehat{m}(d) \;=\; \frac{\sum_i K_h(D_i - d) \cdot \Delta\tilde{Y}_i}{\sum_i K_h(D_i - d)}$$

- K is a weight function (Gaussian, Epanechnikov, . . .).
- Bandwidth h controls smoothness. Small h = local, noisy. Large h = smooth, biased.

Splines fit globally with K knots. Kernels fit locally with a bandwidth. Same goal.

Bandwidth controls smoothness



Nadaraya−Watson kernel regression on the same simulated data as the spline figures.
Small bandwidth −> noisy fit. Large bandwidth −> oversmoothed. The trick is picking h.

Same simulated data as the spline figure. Three bandwidths. The trick is picking h.

Fitting kernel regression in Stata: `lpoly`

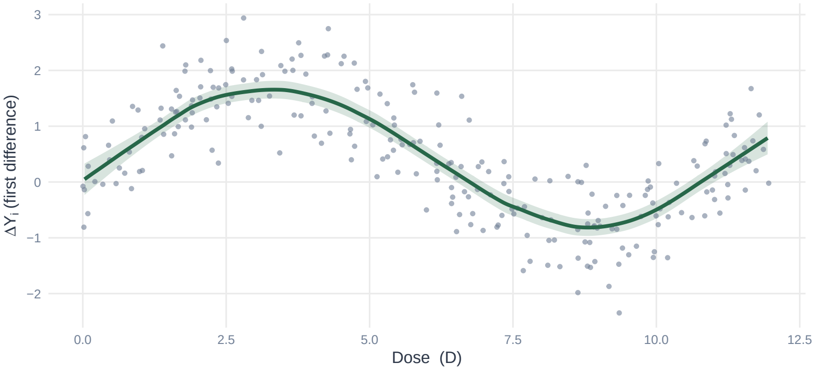
`lpoly` fits a local polynomial; `degree(0)` gives Nadaraya–Watson.

```
* Local-linear (degree 1) kernel fit on a grid of doses
lpoly delta_y dose, kernel(epanechnikov) degree(1) ///
      bwidth(1.2) generate(dgrid yhat) se(se_yhat)

* Plot the fit with a 95% pointwise band
gen lo = yhat - 1.96*se_yhat
gen hi = yhat + 1.96*se_yhat
twoway (rarea lo hi dgrid, color(green%20)) ///
       (scatter delta_y dose, mcolor(gs10) msize(small)) ///
       (line yhat dgrid, sort lcolor(dkgreen) lwidth(medthick)), ///
       xtitle("Dose") ytitle("First difference") legend(off)
```

`bwidth(.)` picks the bandwidth by rule-of-thumb. Use cross-validation for a data-driven choice.

What the kernel regression estimates



Local−linear kernel fit (green line) with 95% pointwise confidence band.
Bandwidth chosen by cross−validation. Same data as the spline figure for direct comparison.

Local-linear kernel fit with 95% pointwise band. Compare to the spline fit on the same data — the curves are nearly indistinguishable.

Where does the spline machinery come from?

Estimating ATT$(d \mid d)$ as a smooth function of d is a classical nonparametric problem. The tools are not new.

Sieve regression (Grenander 1981; Chen 2007 *Handbook*). Approximate an unknown function by a finite-dimensional basis — B-splines, polynomials, wavelets — and let the basis grow with the sample. Bias falls as you add terms; variance is controlled by how fast.

Uniform inference (Chen & Christensen 2018 *QE*; Chen, Christensen & Kankanala 2024). Honest *confidence bands* that cover the whole curve at once — so you can say "the dose-response is positive on $[d_1, d_2]$."

What's new in CBS isn't the estimator. It's the identification.

CBS use B-spline sieve regression

The `contdid` package implements CBS's recipe. It wraps Chen–Christensen–Kankanala's `npiv`.

Three steps:

1. Center the outcome: $\Delta\tilde{Y}_i = \Delta Y_i - \widehat{m}(0)$, where $\widehat{m}(0)$ is the average ΔY for untreated units.
2. B-spline regression of $\Delta\tilde{Y}_i$ on $\psi(D_i)$, using only treated units ($D_i > 0$). Here $\psi(d) = (\psi_1(d), \ldots, \psi_K(d))$ is the *B-spline basis* — K pre-computed columns, one per basis function. In Stata: `mkspline` builds it.
3. Bootstrap to get pointwise confidence bands on the fitted curve.

Returns $\widehat{\text{ATT}}(d \mid d)$ as a continuous function of d, with standard errors at every d.

What we have to assume

- **Strong parallel trends.** $E[\Delta Y(0) \mid D = d] = E[\Delta Y(0) \mid D = 0]$ for *every* d — not just $d = 0$ vs. $d > 0$. Rules out selection-on-dose-response.
- **Smoothness of ATT$(d \mid d)$.** Continuous, bounded derivative in d. The sieve approximates a smooth function, not arbitrary jumps.
- **Overlap.** Enough observations near every d where the curve is estimated.
- **Tuning chosen sensibly.** Number of knots from cross-validation or a rule of thumb.

Strong PT is the load-bearing assumption. The rest are technical.

The estimator, formally

Let $\psi(d) = (\psi_1(d), \ldots, \psi_K(d))^\top$ be the cubic B-spline basis at dose d.

Fit OLS on the treated units:

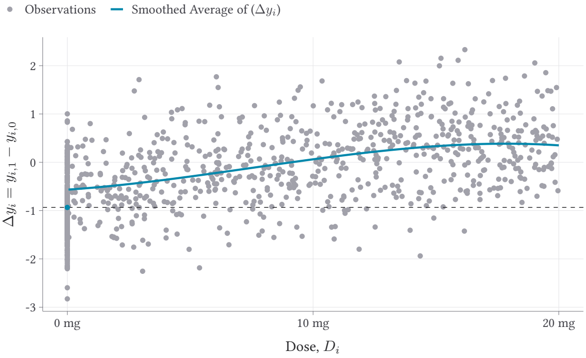
$$\widehat{\theta} \;=\; \arg\min_{\theta} \sum_{i:D_i>0} \Big(\Delta\tilde{Y}_i - \theta^\top \psi(D_i)\Big)^2$$

The dose-response curve and its derivative:

$$\widehat{\text{ATT}}(d \mid d) \;=\; \widehat{\theta}^\top \psi(d) \qquad \widehat{\text{ACRT}}(d) \;=\; \widehat{\theta}^\top \psi'(d)$$

One regression delivers both target parameters.

Option 3: Non-parametric smoothing



A continuous $\widehat{ATT}(\cdot)$ over the dose range, with pointwise confidence bands.

The `contdid` package

Brantly Callaway's R package `contdid` implements all three options.

```
library(contdid)
result <- cont_did(
  yname = "outcome", tname = "year", idname = "unit_id",
  dname = "dose", gname = "treatment_timing",
  data = panel_data,
  target_parameter = "level", # "level" = ATT(d|d); "derivative" = ACRT
  aggregation = "dose", # or "overall"
  control_group = "notyettreated"
)
summary(result)
ggcont_did(result, type = "att")
```

`"derivative"` requires strong PT; `"level"` only standard PT.

Pros and cons of non-parametric smoothing

Pros.

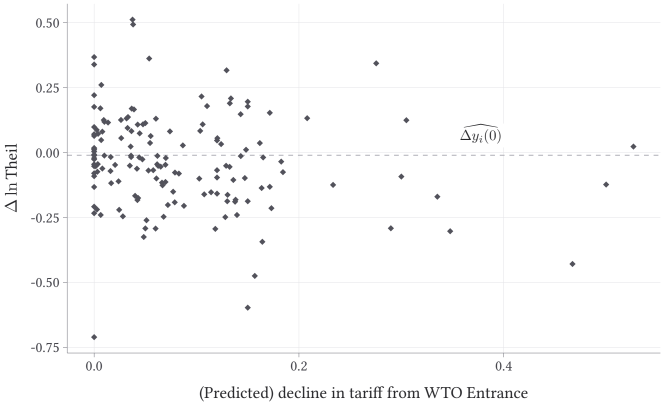
- Returns a continuous function, not a step.
- Confidence bands.
- Less ad-hoc than bin choice.

Cons.

- Bandwidth/smoothing choice is the new discretion.
- Curse of dimensionality at the tails of the dose distribution.
- Harder to communicate to a non-technical audience.

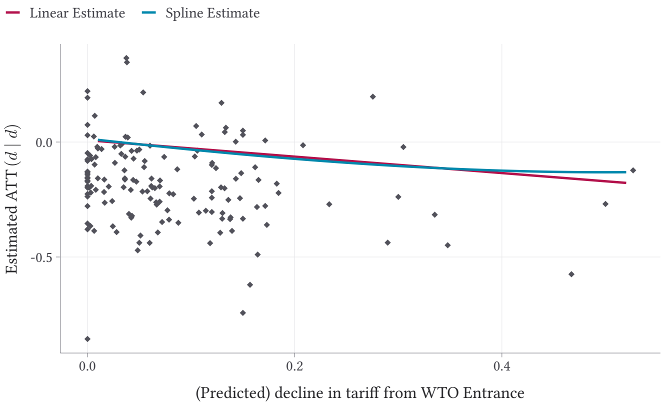
In practice: bin first, smooth second, report both.

The WTO example, done properly



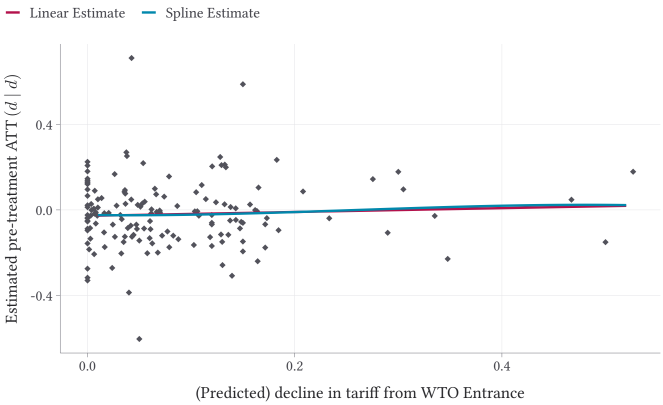
Raw outcome trajectories by tariff-cut group; the setup before any DiD.

The dose-response estimates



$\widehat{ATT}(d \mid d)$ across the dose distribution. A real curve, not the single -0.322 that TWFE delivers.

Pre-trends, dose by dose



Pre-period placebos: should be near zero under standard PT.

Closing: what to do tomorrow

Don't run TWFE with a continuous dose and report $\widehat{\beta}$ as "the dose effect." Under heterogeneity, $\widehat{\beta}^{\text{twfe}}$ is a weighted average with sign-flipping weights, scaled by features of the dose distribution that have nothing to do with the causal question.

Do:

- Pick a target parameter: **ATT($d \mid d$)** or **ACRT(d)**.
- Estimate that parameter: overall, bins, or non-parametric.
- State the identifying assumption: standard PT or strong PT.
- Report pre-trend placebos at each dose.

The continuous DiD machinery is now available — and the cost of doing it right is no longer prohibitive.